Quoted Correctly, Extracted Differently: Measuring the Checks Between a Language Model and a Knowledge Graph on Hadith Narrator Chains

Yahya Efe Kuruçay, Deniz Büyükşahin, Alperen Ulukaya
Department of Computer Engineering, Akdeniz University, Antalya, Türkiye
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Abstract. Systems that build knowledge graphs with large language models commonly ask the model for a source quotation and a confidence score, and use them to decide which extracted facts to trust. We test that practice inside a working extraction engine on the narrator chains (isnad) of Arabic hadith texts. The engine treats the model as a proposer: every candidate carries a quotation, deterministic checks route doubtful candidates to human review, and each accepted fact has a recorded acceptance. On 300 hadiths from the six canonical collections, scored against platform-derived narrator labels, a model given only a schema returns quotations that are all found in the source, so a quotation check flags nothing. A check that the extracted name occurs inside the model's own quotation flags 9.2% of its proposals, and these contain 122 of the 145 proposals that do not match the reference. Most of them are not invented narrators: the model transliterated the name, changed its grammatical case, or replaced a kinship reference by a name. At a comparable share of proposals sent to review, the model's confidence finds 50 unmatched proposals where the check finds 122. Adding type descriptions and a few domain hints to the prompt reduces unmatched proposals from 145 to 5 and leaves the checks little to find. A rule-based parser without any model call is competitive under loose matching and clearly worse under strict matching. The engine stores a person's decisions and derives everything else again; in a simulation on 27,056 labelled chains, an identity decision keyed by the written name alone is reused wrongly for one mention in ten, and narrowing its scope when a deterministic rule sees a contradiction brings that to 0.014 with about half the reviewer questions of a fixed narrow key. The reference labels are not a human gold standard, no reviewer took part in the runs, and one model produced usable results; we state what follows from the measurements and what does not.

1 Introduction

Extraction with large language models has made it cheap to turn text into graph facts. A model receives a passage and a schema and returns nodes and relations. The difficulty is not producing facts but deciding which of them may enter a graph that people will rely on. Two safeguards are common. The model is asked to quote the source for each fact, and it is asked how confident it is. A quotation is meant to show where a fact came from; a confidence score is meant to select the facts a person should check.

We examine both safeguards in a setting where a wrong fact has consequences and where labelled data exists. A hadith opens with its isnad, the chain of people through whom the report was transmitted: "A told us, from B, from C". Scholars judge a hadith from the people in its chain, so one wrong name changes the judgement. Fine-tuned models already extract narrators and their links well when large labelled training sets are available [1], a knowledge graph of the six canonical collections exists [2, 3], and narrator names have been linked to biographical databases [4]. We do not try to improve on that work. We ask a different question: when a general-purpose model proposes facts, which checks between the model and the graph find the proposals that depart from the text, and what do they cost?

The engine we use is built on a division between two kinds of fact. A candidate that a parser or a model extracts from a text can be extracted again, so it is treated as derived and is checked, not trusted. A person's acceptance, correction or identification of a narrator cannot be derived from the text and is stored once. Most of this paper is about checking the first kind. Section 5.7 reports a simulation about the second: what storing an identity decision buys, and what its reuse costs.

Our contributions are the following.

- An engine in which a model can only propose, deterministic checks route candidates to review with a reason code, and every accepted fact has an acceptance record that names a reviewer or an automatic policy (Section 3).
- A measurement, made inside that engine on 300 unseen hadiths, of what two inexpensive checks flag: quotation-in-source and value-in-quotation. The first flags nothing. The second flags 84% of the proposals that do not match the reference labels (Section 5.2).
- A classification of what the flagged proposals are. Under these labels most "errors" of a schema-only model are unrecorded transformations of a correct name, not fabricated narrators.
- A comparison of the check with the model's confidence at equal review budgets (Section 5.3), and of four extraction conditions run through one pipeline with paired bootstrap intervals (Section 5.1).
- A baseline that quantifies how often a written name stands for more than one narrator in labelled chains (Section 5.6).
- A simulation of a ledger of identity decisions on two labelled corpora, in which the scope of a stored decision is derived from contradictions that a deterministic rule detects (Section 5.7).

2 Related work

Hadith corpora and extraction. Sanadset 650K provides 650,986 records with narrator tags produced by regular expressions and text matching; its data paper reports no accuracy measurement [5]. The Ifta Sunnah data family restructures the markup of an official hadith platform into narrator spans, transmission relations and narrator identifiers, and fine-tuned models trained on it reach, according to the author's model repository, F1 0.98 for narrators and 0.95 for relation pairs [1]. Earlier work located isnad spans in long unpunctuated texts and showed how much the matching rule matters: the same CRF output scored F1 0.254 under exact and 0.835 under partial matching, and experts agreed exactly on the end of an isnad in 53% of cases [6]. Segmentation of isnad from report text with n-grams reached 92.5% on 240 hadiths [7].

Narrator identity. AR-Sanad 280K frames narrator disambiguation as classification over 18,298 narrators. On 27,056 real chains its best model reaches micro F1 83.5 while 60.6% of chains contain at least one wrong narrator [8]. A recent pipeline links 185,216 name variants to two biographical databases without an evaluation against labelled links [4].

Hadith knowledge graphs and LLMs. SemanticHadith publishes an ontology and graph for the six collections; its extension extracts entities with a custom NER model and relies on experts checking a sample of 100 instances per collection, and it lists a framework for expert validation of ambiguous names as future work [3]. LLM-based processing of hadith corpora has been evaluated with expert scores on a 0 to 10 scale [9] or by one author comparing 100 records with the source [10]. A critical review names weak benchmark comparability and sparse expert validation as open problems of the field [11].

Checking extracted facts. Text2KGBench measures, besides precision and recall against reference triples, whether the subject and object of a triple occur in the sentence, and reports subject hallucination of 0.17 next to precision of 0.38 for one model and dataset: presence in the text does not imply correctness [12]. Attribution and factuality metrics for generated text work at the level of sentences or atomic claims [13, 14]. Self-reported confidence of language models has been studied as a calibration problem [15, 16]. We did not find a study that measures, for knowledge graph extraction, the separate effect of a check on the relation between a model's quotation and the value it extracted. Our search was not systematic.

3 System

GiraffeGraph Core is a schema-first extraction engine with a FastAPI backend, PostgreSQL for metadata, Neo4j for the committed graph and a web dashboard. A project defines node types, relation types, their properties and the properties that form the identity of a node or relation. A source text is ingested as a source unit and cut into segments. A provider, which may be a language model or a rule-based program, proposes candidate nodes and relations for the segments; each candidate carries one or more quotations with a segment reference. Validation then assigns each candidate a status: validated, needs review, or rejected. A run can be committed to the graph only when no candidate of the run waits for review.

Checks. Three checks concern us here. (i) *Schema*: a candidate whose type is not in the schema is dropped on its own, the other candidates of the call survive, and the unknown type is recorded as a review item; undeclared property keys are removed and recorded the same way. (ii) *Quotation in source*: the candidate's quotation must occur in the text of the segment it cites. (iii) *Value in quotation*: a string value that forms the identity of a node, such as a narrator's name, must occur in one of the candidate's quotations. Comparisons ignore Arabic vowel marks and letter variants. A candidate that fails check (ii) or (iii) is routed to review whatever its confidence, and is not rejected, because a failed check does not show that the fact is false.

Acceptance record. A review decision is stored with the reviewer's name, the time, the time spent and, for an edit, the value before and after. Decisions are appended, never overwritten, and later re-validation does not change a candidate a reviewer has decided. For each committed fact the engine reports either the human decision or the automatic policy that accepted it.

Relation identity. A relation type can declare identity properties. For the chain relation these are the hadith reference, the chain number and the position, so that two transmissions between the same two narrators in different hadiths are two edges. Without this, the second transmission overwrote the first.

Case-study components. Three components are specific to hadith: a rule-based segmenter that finds the chain span from a lexicon of 26 transmission formulas and splits it at the chain-switch marker, a schema template (narrator, hadith, "narrated from", "transmits") with short extraction hints, and a provider that turns the segmenter's output into candidates without any model call.

4 Experimental setup

Data. The test set contains 300 hadiths, 50 from each of the six canonical collections, drawn with a fixed seed from the Ifta data after removing every record whose text matches one of 50 hadiths used during development [17]. The reference labels are the narrator mentions marked on the platform, 1,778 in total, between 2 and 11 per hadith (median 6); 74 hadiths contain more than one chain. The labels are platform markup restructured by a third party, not annotations by independent annotators, and they are inconsistent in places: narrators inside a compiler's comment are sometimes marked and

sometimes not. For 253 of the 300 hadiths a matching record exists in Sanadset; its regular-expression labels agree with the Ifta labels at F1 0.959 under loose matching (95% interval 0.943 to 0.972).

Scoring. The unit of analysis is the hadith. System and reference mentions are aligned by maximum-weight bipartite matching. Under *strict* matching two mentions match when their normalised strings are equal; under *loose* matching also when one contains the other or their token F1 is at least 0.60. Precision, recall and F1 are micro-averaged. Intervals are 95% percentile bootstrap intervals over 2,000 resamples of hadiths, and differences between conditions use paired resamples. A proposal is called *unmatched* when it is not aligned to a reference mention under loose matching. Unmatched is a statement about the labels' surface form, not about truth.

Conditions. All conditions run through the same engine with no review decision.

Condition	Input to the provider	Provider	Prompt
Rule	Chain span from the segmenter	rule-based, no model	none
Schema only	Whole hadith, one call	Claude Haiku 5.5	Type names and output contract; descriptions removed
Isnad-aware	Whole hadith, one call	Claude Haiku 5.5	Descriptions and hints: copy names verbatim, exclude transmission verbs, the Prophet is not a narrator
Isnad-aware, chain span	Chain span from the segmenter	Claude Haiku 5.5	Same as isnad-aware

The model runs with structured output and its effort parameter set to `low`. Responses are stored in a content-addressed cache, so every run can be replayed without new calls. The runs used the engine's in-memory storage; the same pipeline is covered on PostgreSQL and Neo4j by integration tests.

5 Results

5.1 Extraction

Table 1. Narrator mentions on 300 hadiths (1,778 reference mentions). P, R, F1 under loose matching, with the 95% interval of F1.

Condition	P	R	F1	Strict F1	Exact chains
Rule	0.968	0.881	0.923 [0.910, 0.935]	0.633	0.490
Schema only	0.914	0.868	0.890 [0.865, 0.914]	0.826	0.567
Isnad-aware	0.997	0.919	0.956 [0.946, 0.966]	0.904	0.690
Isnad-aware, chain span	0.998	0.903	0.948 [0.936, 0.959]	0.897	0.650

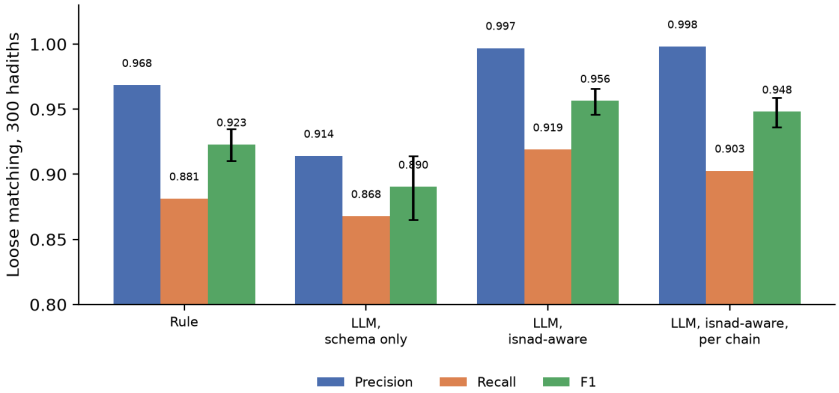


Figure 1. Loose precision, recall and F1 by condition. Error bars on F1 are 95% bootstrap intervals.

Under loose matching the schema-only model is below the rule-based parser by 0.032 F1 (interval 0.006 to 0.060). Under strict matching the order is reversed, 0.826 against 0.633: the parser finds the right names but its spans run past the end of a name, while the model returns clean names and alters some. Adding descriptions and hints to the schema-only prompt raises F1 by 0.066 (0.041 to 0.091) and precision by 0.083 (0.059 to 0.108); unmatched proposals fall from 145 to 5. Restricting the model's input to the chain span does not help (difference 0.008, interval −0.002 to 0.019). Only 10 of the 300 hadiths were split into more than one call, so this condition does not test per-chain calling on long hadiths. All conditions lose recall on hadiths with parallel chains; the parser loses most (0.924 on single-chain hadiths, 0.778 otherwise).

5.2 What the checks flag

In the schema-only condition every quotation that reached the check was found in its segment. The quotation check flagged no candidate. The value-in-quotation check flagged 155 of 1,688 proposed mentions (9.2%).

Table 2. Schema-only condition: proposals by match with the reference and by the value-in-quotation check.

	Unmatched	**Matched**
Flagged	122	33
Not flagged	23	1,510

Of the flagged proposals 79% are unmatched, and they are 84% of all 145 unmatched proposals. Table 3 classifies the flagged names with fixed rules.

Table 3. Flagged names by kind of change (schema-only condition).

Kind of change	**Example: quotation, then name**	**Flagged**	**Unmatched**	**Matched**
Transliterated into Latin script	حَدَّثَنَا الْأَعْمَشُ, "al-A'mash"	79	79	0
Grammatical case changed	أَبُو, عَنْ أَبِي بِشْرٍ بِشْرٍ	34	21	13
Kinship reference replaced by a name	عَنْ أَبِيهِ ("from his father"), عُرْوَةَ	25	18	7

Kind of change	Example: quotation, then name	Flagged	Unmatched	Matched
Other, name elsewhere in the text		7	3	4
Other, name not in the text		10	1	9

Three points qualify these numbers. First, most unmatched proposals name the right narrator. Seventy-nine are Latin transliterations, all from 14 hadiths in which the model switched script, and 21 are a change of grammatical case. The same case change is unmatched on a two-word name and matched on a longer one, because of the token threshold of the loose rule; the check and the scorer both compare surface forms, so part of their agreement is by construction. Second, the kinship cases are inferences the text supports: in the example the text reads "Hisham ibn Urwa, from his father". Fifteen of the 25 names occur in the hadith text. We found one flagged name that the text does not contain in any form. Third, the check does not cover category errors. Ten unmatched proposals name the Prophet as a narrator; the check flags the two written in Latin script and misses the eight that are copied correctly.

What the check establishes is that the model wrote something other than what it quoted and did not say so. Of the 145 unmatched proposals, 102 have a name that does not occur in the hadith text, and the check flags all 102.

Because a flagged candidate is not accepted until a person decides, candidates accepted without review have precision 0.985 instead of 0.914, at recall 0.849 instead of 0.868. The graph receives less until a reviewer acts: a run with an open review item is not committed, which holds back 110 of the 300 hadiths and leaves the committed output at precision 0.983 and recall 0.582.

In the isnad-aware condition 5 of 1,639 proposals are unmatched and the check flags 3 of them, with no matched proposal flagged.

5.3 Confidence against the check

The area under the ROC curve of the model's confidence as a predictor of a matched proposal is 0.634 in the schema-only condition. The model used seven distinct values, and 1,623 of 1,688 proposals have a confidence between 0.8 and 1.0.

Table 4. Unmatched proposals sent to review by a confidence threshold and by the check (145 unmatched in total).

Signal	Proposals sent to review	Unmatched among them
Confidence below 0.80	65 (3.9%)	16
Confidence below 0.85	275 (16.3%)	50
Confidence below 0.90	1,018 (60.3%)	110
Value-in-quotation check	155 (9.2%)	122

5.4 Review workload and cost

The schema-only condition sends 538 candidates to review (155 nodes and 383 relations) and commits 190 of 300 hadiths without review. The isnad-aware condition sends 16 and commits 294. With the provider's list prices of 6 October 2026, extraction costs USD 1.87 per 1,000 hadiths in the schema-only condition and USD 2.29 in the isnad-aware condition, at a median latency of 11 to 13 seconds per call. All runs in this paper cost USD 3.49. We did not measure human review time.

5.5 A failed run with a second model

We ran a larger model of the same family, Claude Sonnet 5.5, with the same settings on 36 of the hadiths. In the isnad-aware condition it returned a single narrator for 28 of them and usually wrote the narrator's name into the field for a property's key. Loose recall was 0.237 against 0.916 for Haiku 5.5 on the same hadiths. We suspect two causes in our configuration, an ambiguous wire format in which both the list entry and the property are called `name`, and a reasoning setting that is not calibrated alike across models; we did not test either. The run is not a comparison of models. It does show the engine failing safely: 29 unknown property keys were recorded for review, nodes without an identity value were rejected, 30 of 36 runs were blocked from commit, and the precision of what was proposed stayed at 0.966 or above.

5.6 A written name is not an identity

The engine identifies a narrator node by the written name. The Ifta labels attach a person identifier to each chain position, which lets us measure the consequence without running a system. The 300 hadiths have 2,150 chain positions filled by 740 people. Grouping positions by the name exactly as written gives 964 groups; 24 of them cover more than one person and contain 10.0% of the positions, and 25.8% of the people are split over several groups (pairwise precision 0.737, recall 0.559; B-cubed 0.941 and 0.715). Normalising the name reduces splitting (21.8%) and increases conflation (12.2% of positions). The most ambiguous name is a kinship reference, "his father", which stands for 34 people.

5.7 Storing the identity decision

If the name is not the identity, a person has to say who is meant, and that decision is worth keeping. We simulated a ledger that reads labelled chains one mention at a time, asks a reviewer when it meets a key it has not seen, and reuses the answer afterwards. The reviewer's answer is the label of the dataset. A question is anything that goes to the reviewer; a silent error is a reused answer that disagrees with the label. The data are the 27,056 test chains of AR-Sanad 280K (150,904 mentions, 6,378 people) [8] and, as a replication on labels from another source, the 34,681 hadiths of the six Ifta collections that carry narrator identifiers (205,514 mentions). Values are means over five random orders.

What matters is the scope of a stored decision: how much of the context a later mention must share before the answer is reused. In the narrowing policy every name starts with the name alone and is tied to the narrator named before it only after a contradiction has been seen for that name. The detector is a rule on the narrator catalogue that asks the reviewer when the proposed person and the neighbouring narrator are too many generations or too many years of death apart; its bounds were chosen on AR-Sanad and applied to the Ifta data unchanged.

Table 5. Questions and silent errors per mention.

Policy	AR-Sanad: questions	AR-Sanad: silent errors	Ifta: questions	Ifta: silent errors
Fixed scope: name	0.101	0.100	0.077	0.102
Fixed scope: name; the rule asks about the flagged mention only	0.204	0.054	0.229	0.049
Scope narrowed by the rule	0.178	0.014	0.199	0.010

Policy	AR-Sanad: questions	AR-Sanad: silent errors	Ifta: questions	Ifta: silent errors
Fixed scope: preceding narrator and name	0.340	0.006	0.292	0.007

Keyed by the name alone, a decision is reused wrongly for one mention in ten. Narrowing the scope where the rule sees a contradiction comes close to the error of the narrow key with about half to two thirds of its questions. The same rule used only to send the flagged mention to the reviewer leaves the error near 0.05: what helps is that one contradiction changes how every later occurrence of the name is handled. With the ledger filled from five Ifta collections and counted on the sixth, the narrowing policy had between 0.004 and 0.016 silent errors per mention and asked fewer questions than the narrow key in each of the six collections.

A stored decision must also be found again after the text is extracted a second time. With the pair of hadith and normalised name as the target, between 0.857 and 0.977 of the narrator candidates of one model run have a counterpart in a run with another prompt, and between 0.561 and 0.634 between the rule-based parser and a model run.

6 Discussion

The quotation is not where the model departs from the text. A quotation requirement is satisfied by pointing at real text. In our runs the model did that every time and then wrote a value that the quoted text does not contain. A check on the quotation alone therefore says nothing about the value. Checking the value against the quotation costs one more string comparison and was the only signal that separated matched from unmatched proposals.

Flagged is not false. One flagged name in five matched the reference, and most unmatched ones were the right person in another form. A check of this kind should route to a person or to an explicit rule, not reject. A rule that accepts changes of grammatical case and records that it did so would clear 13 of the 33 matched names that were flagged.

Instructions matter more than checks, and checks matter when instructions fail. A few short hints and the type descriptions removed almost all unmatched proposals, more than any check found. The checks earn their place in the cases where the prompt, the model or the configuration is wrong, as the failed run illustrates.

Strict acceptance moves work to the reviewer. Keeping a whole hadith out of the graph while one candidate waits for review protects the graph and leaves it incomplete: recall 0.582 in the schema-only condition before any review. The design is only as good as the review step, which we have not measured.

What is kept is the decision. The candidates of an extraction are replaceable; the same text can be read again with another prompt or another model. The identity of a narrator and the acceptance of a fact are not in the text. The simulation suggests that such decisions are reused often, that reuse by name alone is unsafe, and that the scope of a decision can be left broad until a cheap rule finds a contradiction. The engine does not yet do this: it attaches a decision to a candidate, and it has no date rule.

7 Limitations

The reference is platform markup, not a gold standard produced by independent annotators, and absolute scores should not be read as accuracy. The share of unmatched proposals that the value-in-quotation check flags is specific to this task and prompt: on 300 abstracts of the human-annotated BC5CDR corpus [18], with the same model and a schema-only prompt, the check flagged 3.9% of 2,269 proposals and these held 53 of the 303 that did not match the annotations, while the model's confidence found 58 at the same review share; most unmatched proposals there were verbatim copies of spans the annotators had not labelled. No reviewer took part, so the human part of the loop is implemented and tested but not evaluated. In the ledger simulation the reviewer is the label of the dataset: the first answer is free and always right, a question counts as one unit whatever time it would take, and the six collections share most of their narrators. All usable results come from one model, one setting and one run per condition. An unmatched proposal is not necessarily a false statement. Narrator mentions were scored, not relations. The hadiths are well known and are probably in the model's training data; we did not test memorisation with altered names. Schema drift, which motivated the type and property checks, was almost absent with the model that worked, so its frequency in normal use is not established. The rule-based segmenter was tuned on 50 hadiths that are excluded from the test set.

8 Conclusion

Inside a working extraction engine and on 300 unseen hadiths, a check that the extracted value occurs in the model's own quotation flagged most proposals of a weakly instructed model that do not match the reference, while a check that the quotation occurs in the source flagged none and the model's confidence found far fewer at the same review budget. What the check finds are mostly unrecorded transformations of correct names. Better instructions made the problem nearly disappear for this model. In a simulation, a stored identity decision whose scope narrows on contradiction came close to the error of a narrow key with fewer questions. The next steps are a human-labelled reference, a reviewer in the loop with measured review time, the ledger repeated with that reviewer's decisions, more models, relation-level scoring, and linking narrator names to an authority list.

Statement on the use of AI tools

AI assistants wrote most of the code of the evaluated system, the analysis scripts and a first draft of this text under the authors' direction. Bibliographic records were retrieved from indexes or publishers' pages. Every measured number is produced by a script from stored model responses. A draft was checked against the result files by an independent AI reviewer, and the claims were narrowed where the files did not support them. The authors are responsible for the content.

Data and code

The code, the evaluation scripts, the cached model responses and the result files are in the project repository, which is private while the project is in development; we plan to publish the engine under an open-source licence at the end of the project. The hadith data comes from third-party sources and is used for evaluation only; the manifest lists each source with the licence statement as given by the source [17].

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